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Authors: Giltz, Sarah M., and Taylor, Caz M.

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REDUCED GROWTH AND SURVIVAL IN THE LARVAL BLUE CRAB *CALLINECTES SAPIDUS* UNDER PREDICTED OCEAN ACIDIFICATION

SARAH M. GILTZ* AND CAZ M. TAYLOR

Department of Ecology and Evolutionary Biology, Tulane University, 6823 St. Charles Avenue, New Orleans, LA 70118

ABSTRACT The decreasing average pH of the world's oceans, called ocean acidification, can be detrimental to marine organisms. Decreased pH causes dissolution of calcium carbonate shells, as well as changes to metabolism, behavior, growth, survival, and early development of many species; however, the impacts of lower pH on crustaceans have not been widely researched. This study investigated the effects of lower pH on the growth and survival of larval blue crabs *Callinectes sapidus* through laboratory experiments. Blue crab larvae live at the surface of the open ocean where the pH is decreasing with ocean acidification. A pH of 7.8 was used to mimic the average surface ocean pH predicted to be attained by 2,100 by the Intergovernmental Panel on Climate Change and was compared with a control pH of 8.2, the current average surface ocean pH. Larvae raised at the lower pH were 10% smaller and there was a 23% reduction in survival. Larval populations within an aquarium with a lower pH were more likely to exhibit sudden die-offs than larvae raised at the control pH.

KEY WORDS: *Callinectes sapidus*, ocean acidification, crabs, climate change, larvae, Gulf of Mexico, crustaceans, molting

INTRODUCTION

Impacts of global climate change include increasing ocean temperatures, rising sea level, and a decrease in average ocean pH (Orr et al. 2005, Feely et al. 2009). The decrease in pH, called ocean acidification, is caused by burning of fossil fuels, which leads to more carbon entering the oceans initiating chemical reactions that reduce seawater pH (Sabine et al. 2004, Riebesell 2008). The average global ocean pH was 8.2 before the industrial revolution, has since dropped to 8.1 (from about 1,760 to 2,010), and is predicted to drop further to 7.8 by 2,100 (Malakoff 2012). If the predictions hold, this represents a 150% increase in the acidity of the world's oceans in under 300 y (Caldeira & Wickett 2003, Raven et al. 2005, Doney et al. 2009, Feely et al. 2009).

The long-term impact of ocean acidification on marine ecosystems remains to be determined but the immediate effects of reduced pH have been well studied for some organisms (Takahashi 2004, Malakoff 2012). The growing body of literature on ocean acidification shows a variety of species-specific responses to a lowered pH. The best-documented effect is dissolution of calcium carbonate leading to weakened shells, particularly in molluscs (Feely et al. 2004, Orr et al. 2005, Przeslawski et al. 2008). Lowered pH can also cause reduction in reproduction, growth, and survival (Doney et al. 2009, Dupont & Thorndyke 2009, Beniaish et al. 2010, Weydmann et al. 2012). Effects of ocean acidification can vary by life history stage (Dupont & Thorndyke 2009) and are not exclusively negative; lowered pH has been shown to increase growth rate in an echinoderm (Gooding et al. 2009) and to increase photosynthesis (Riebesell 2008, Doney et al. 2009). In laboratory studies with marine invertebrates, lowered pH caused changes in fertilization, hatching success, foraging behavior, and early development (Kurihara & Shirayama 2004, Weydmann et al. 2012, Dodd et al. 2015) as well as slowed development

(Dupont et al. 2008) and increased mortality (Wood et al. 2015, McLaskey et al. 2016). The focus of ocean acidification research is moving from the adult to larval stages and increasingly covering taxa beyond molluscs and echinoderms, first identified for their greater vulnerability (Whiteley 2011).

There are relatively few ocean acidification studies on crustaceans, although this taxon may be highly susceptible to lowered pH because of frequent molting, pH-dependent cuticle formation (Neufeld & Cameron 1992, Egilsdottir et al. 2009), and the presence of calcium carbonate in the carapace (Vigh & Dendinger 1982). Crustaceans regulate their internal acid–base balance in response to lowered environmental pH and show an increased $[\text{HCO}_3^-]$ in the hemolymph (Cameron 1978, Cameron & Iwama 1987). Laboratory experiments with crustaceans have shown reduced growth in larval lobsters *Homarus gammarus* (Arnold et al. 2009, Small et al. 2016) and amphipods *Echinogammarus marinus* (Egilsdottir et al. 2009). Larval mortality increased with lowered pH in krill *Euphausia pacifica* (McLaskey et al. 2016). In ocean acidification studies on crab species, lowered pH caused decreased growth and fitness in the larval spider crab *Hyas araneus* (Walther et al. 2010), decreased growth as well as decreased survival in juvenile red king crabs *Paralithodes camtschaticus* and juvenile tanner crabs *Chionoecetes bairdi* (Long et al. 2013). Most studies of ocean acidification have focused on the juvenile or adult stage but larval stages may be more susceptible to added stress, potentially affecting recruitment into the adult population (Byrne 2011, Wood et al. 2015). The larval stages of many marine invertebrates are more vulnerable to stress than adults and will encounter the open ocean pH decrease (Dupont et al. 2008, Whiteley 2011). Studies are lacking on commercially important species although many fishery species live in the open ocean or have a pelagic larval stage that faces a decreasing pH (Hall-Spencer et al. 2015). Understanding the potential impacts of increasing ocean acidification is crucial for the continued security of marine organisms and fisheries that depend on them.

The blue crab *Callinectes sapidus* is generally a well-researched fishery species but not with regard to ocean acidification. A small number of studies have been conducted on

*Corresponding author. E-mail: sgiltz@tulane.edu
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juvenile or adult stages (Ries et al. 2009, Glandon & Miller 2016) but no studies have been conducted on larval stages. The blue crab constitutes one of the most valuable crab fisheries in the world (Eggleston et al. 2008). Blue crabs have a significant ecological role, influencing the estuarine carbon cycle and serving as a potential keystone species in marsh environments (Baird & Ulanowicz 1989, Silliman & Bertness 2002). Juvenile and adult crabs are morphologically distinct from larvae and live in a different environment. For the first month after being spawned offshore, blue crab larvae (zoea) remain in the pelagic environment, moving passively by ocean currents thereby making them susceptible to changes in open ocean pH. Larval crustaceans molt more frequently than juveniles and adults and thus require more energy to regularly rebuild an exoskeleton (Arnold et al. 2009). The cuticle layer of the blue crab carapace is 22% calcium, making the crabs potentially sensitive to changes in pH that may disrupt calcium absorption (Vigh & Dendinger 1982). At the juvenile stage, blue crabs transition from the open water to coastal habitats. Near shore habitats tend to experience larger pH fluctuations adding further complexity to the potential threat of ocean acidification (Wahl et al. 2016). As estuarine organisms, blue crab juveniles and adults must be more tolerant of pH fluctuations. In studies of later stage blue crabs, an increase in calcification was observed in adults at high pCO₂ levels (Ries et al. 2009), and no effect was seen on the growth or feeding of juveniles (Glandon & Miller 2016).

The objective of this study was to investigate the effects of ocean acidification on the survival and condition of the larval blue crab. Laboratory experiments were used to explore the hypothesis that the lowered pH environment predicted for the open ocean environment by the end of the 21st century will negatively affect growth and survival.

MATERIALS AND METHODS

Between May and July 2014, four gravid females were caught offshore near Cocodrie, LA, at the Louisiana Universities Marine Consortium (LUMCON) facility (29.26° N, 90.66° W). Each female was kept in outdoor water tables until spawning, and the entire experiment was repeated separately with larvae from each of the four different females to minimize the differences in response based on genetic variation.

After hatching, zoea were randomly placed into treatment groups and moved to aquaria at the assigned pH following an acclimation period. The treatments consisted of ocean acidification (OA) aquaria at pH 7.8 and CONTROL aquaria at 8.2. The pH was adjusted by bubbling compressed CO₂ into filtered seawater to maintain a pH within ± 0.15 pH units of the target value. Air was continuously pumped into all aquaria. Larvae were reared in 10-gallon aquaria containing filtered seawater with water temperatures between 24°C and 27°C (Sulkin & Epifanio 1975, Zmora et al. 2005). Aquaria were monitored daily for ammonia levels, salinity (25–27 ppt), and temperature. Fifty percent water changes were performed whenever NH₃/NH₄⁺ rose above 0.25 mg l⁻¹. Food included live rotifers, brine shrimp, and phytoplankton (*Nannochloropsis* and *Isochrysis*). Rotifers were added daily as needed to establish 10 ml⁻¹, and brine shrimp were added when zoea were large enough to consume them (~day 8).

Methods used in the four iterations of the experiment with different females were identical other than the number of starting zoea, number of aquaria used, and age of zoea (Table 1).

TABLE 1.
Summary of larval survival and average carapace width.

Experiment number*	1		2		3		4		Pooled	
	C†	OA	C	OA	C	OA	C	OA	C	OA
Healthy aquaria (total number aquaria)‡	1 (1)	1 (1)	1 (1)	0 (1)	3 (3)	2 (3)	3 (3)	2 (3)	8 (8)	5 (8)
Percent survival in healthy aquaria	8	10	13	N/A	36	15	29	26	22	17
Average size (mm)	0.651 n = 42	0.592 n = 49	0.697 n = 63	N/A	0.549 n = 640	0.455 n = 182	0.482 n = 519	0.477 n = 306	0.532 n = 1,264	0.478 n = 537

* Spawning dates and age at end of experiment: (1) May 31, 2014, 18 days; (2) June 4, 2014, 19 days; (3) June 29, 2014, 12 days; (4) July 17, 2014, 11 days.

† C, CONTROL; OA, ocean acidification treatment.

‡ Aquaria with population crashes (number of zoea in aquarium decreases within 2 days to under 10 visible individuals) were not used to calculate survival, average size, or for ANOVAs.

The first two experiments were smaller in number of zoea and aquaria, and in the final two experiments, daily size measurements were taken by removing five zoea removed per aquarium per day and measuring the carapace width. In all experiments, the density of zoea was kept very low to avoid crowding (Zmora et al. 2005).

All crabs alive at the end of each experiment were counted and the carapace width recorded using a microscope with reticle. *Survival*, measured per tank, is the number alive at the end of experiment divided by the starting population per tank. *Size* refers to the measurements of individual carapace widths. Analysis of variance (ANOVA) was performed to determine whether the size and survival were significantly different between CONTROL and OA larvae pooled across all experiments. *Replicate* (experiment number) was used as a random effect to account for differences due to genetic variability of different spawns or age or molt stage at the end of experiment. *Aquarium* was also used as a random effect for size ANOVA. Carapace measurements were taken daily from up to five zoea per aquaria in experiments three and four. When aquaria did not have five zoea visible, all visible zoea were measured. *Growth rate* is described by the average daily size measurements per tank over time. A one-way repeated measures ANOVA was conducted to compare the effect of *treatment* (OA versus CONTROL) on growth rate of larvae over the length of the experiment. Repeated measures ANOVA was used because each aquarium was measured each day for the average size of zoea with aquarium and replicate included as random effects.

Larvae were not identified to molt stage during the experiment, thus the mechanisms explaining size and growth rate analyses are confounded. For example, differently sized zoea could be at different molt stages or different sizes within a stage. All zoea were of the same age within each experiment, so any size differences can still be meaningfully observed, but the mechanism cannot be discovered in the study. During the experiments, aquaria were monitored daily for changes. Some aquaria experienced *population crashes* in which over 2 days the number of zoea dipped below 1% of the original number. Survival and size measurements from crashed aquaria were not used in the analyses.

RESULTS

At the end the experiment, larvae in the OA treatment were on average 10% smaller than in the CONTROL treatment (ANOVA, $F_{1,1731} = 46.4$, $P < 0.001$; Table 1; Figure 1) and survival was 23% lower (ANOVA, $F_{1,13} = 4.994$, $P = 0.044$). Larvae in the OA treatment had a significantly slower growth rate than larvae in the CONTROL treatment (repeated measures ANOVA, $F_{1,259} = 16.9$, $P < 0.001$; Figure 2). In total, 1,264 CONTROL larvae were measured and 537 OA larvae. Population crashes were observed in 3 out of the 8 (38%) of the OA aquaria but not in any CONTROL aquaria (Table 1).

DISCUSSION

In this study, larval blue crabs showed diminished growth and decreased survival at a pH that is predicted to occur in the ocean in the next 100 y (IPCC 2007). The results are consistent with other findings in the literature suggesting larval stage invertebrates are vulnerable to lower pH (Dupont et al. 2008).

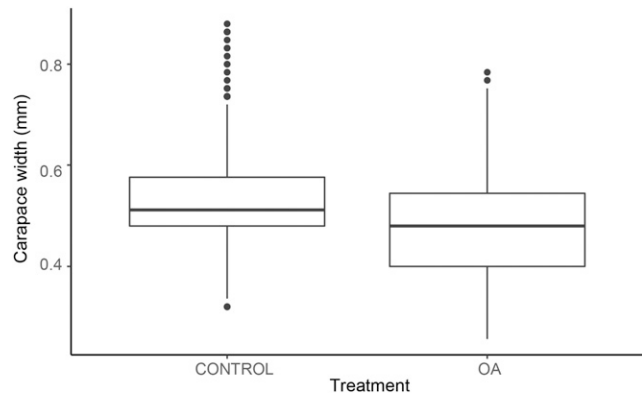


Figure 1. Carapace width measurements of larvae at the end of experiments. The boxes indicate the first and third quartiles. The ends of the whiskers show the data within 1.5 interquartile range of the lower and higher quartile. Larvae in the OA treatment are significantly smaller than CONTROL (ANOVA, $F_{1,1731} = 46.4$, $P < 0.001$). Standard Tukey quartile boxplots are used; CONTROL $n = 1,264$; OA $n = 537$.

Other ocean acidification experiments have shown slowed development in larval and embryonic crustaceans (Arnold et al. 2009, Egilsdottir et al. 2009, Small et al. 2016) as well as juvenile crabs (Long et al. 2013).

The mechanisms behind the suppressed growth and development are not well understood. Whether through decreasing metabolism or increasing energy spent in response to stress, lowered pH may be costly over the long term (Barry et al. 2005, Arnold et al. 2009). Stress response and maintenance of a desired internal pH could divert resources away from molting and growth. Metabolic suppression can occur in marine animals during suboptimal environmental conditions such as hypoxia, suboptimal salinity, or lowered pH, and a shift to chronic lower metabolism may be detrimental (Fabry et al. 2008). Larval sea urchins (*Strongylocentrotus purpuratus*) under high CO_2 were found to be smaller and showed upregulation of genes linked to

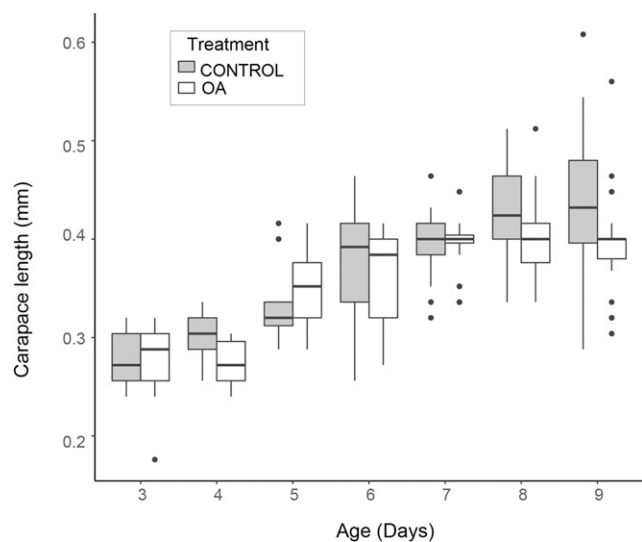


Figure 2. Daily carapace measurements of zoea over. Data combined from zoea from two spawning events. The average size of zoea in the OA and CONTROL treatments differ significantly over time (repeated measures ANOVA, $F_{1,259} = 16.9$, $P < 0.001$).

an increased need for internal pH regulation (Pespeni et al. 2013). Larvae of the blue crab molt more frequently than at later stages, shedding a cuticle and reforming a new one every few days for a month. Unsuccessful molt completion can lead to death and is observed more often in stressful circumstances (Small et al. 2016). Larvae are vulnerable to high predation and death in suboptimal conditions, more so than at later stages (Shanks et al. 2003). Reduction in the size of larvae, time to reach next molt, and increased mortality as seen in this study could lead to larger population level responses and changes in the patterns of larval dispersal (Duarte 2007).

The mortality seen in both control and treatment larvae in this study is high but not inconsistent with other laboratory studies (Sulkin & Epifanio 1975). Generally, the highest mortality occurs in the first and second larval stages, ranging from 53% to 89% during that time under optimal salinity and temperature (Costlow & Bookhout 1959). Studies focused on increasing survival with optimal diet resulted in 17%–30% mortality rates per stage with a cumulative 10% (Sulkin 1975) or 24%–34% (Bookhout & Costlow 1970) survival at the end of the larval period. One large scale hatchery study reached a maximum survival rate of 74%, but averaged 30% survival (Zmora et al. 2005).

Crab larvae raised in a lower pH were more likely to experience an aquarium-wide population crash and exhibited higher mortality even when excluding crashed tanks. Increased risk of aquarium populations crashing in an experimental setup may not be directly applicable to a large, complex natural ecosystem but it could indicate that a biophysical extreme can be reached that puts individuals on the edge of vulnerability. An added stressor (such as an undetected pathogen or contaminant) could lead to mass mortality. More research is needed to understand the reasons for and the wider implications of the crashes.

This study found that ocean acidification as it is predicted to occur over the next century might have a detrimental effect based on the observed 10% decrease in size and 23% reduction in the survival of larval blue crabs. Follow-up studies are needed

to explore the mechanism behind decreased growth and to determine if the difference occurs through less frequent molting or decreased size after molt. More investigations should be made into the increased vulnerability under a stressful pH that could lead to increased mortality as a population. Future studies should also examine the transitions between stages. Decreasing the health of the larval population can diminish the success of later stages (Pechenik 1999).

Effects of ocean acidification have been shown to differ in magnitude and can be negative or positive at different stages in the life cycle (Dupont et al. 2010). The results of this study contrast with the positive or neutral effects of lowered pH on juvenile blue crabs seen in Ries et al. (2009). Results from one stage cannot be extrapolated to give a full picture of the response of the species. Experiments using juveniles or adults alone are not enough. Understanding each life history stage will allow for better informed predictions in the face of climate change and its associated threats. Changes in conditions that lead to developmental differences may have the potential to impact the adult population, which has fishery management and conservation implications (O'Connor et al. 2007). Healthy populations and sustainable fisheries depend on the survival of all stages.

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LITERATURE CITED

- Arnold, K. E., H. S. Findlay, J. I. Spicer, C. L. Daniels & D. Boothroyd. 2009. Effect of CO₂-related acidification on aspects of the larval development of the European lobster, *Homarus gammarus* (L.). *Biogeosciences* 6:1747–1754.
- Baird, D. & R. E. Ulanowicz. 1989. The seasonal dynamics of the Chesapeake Bay ecosystem. *Ecol. Monogr.* 59:329–364.
- Barry, J. P., K. R. Buck, C. Lovera, L. Kuhnz & P. J. Whaling. 2005. Utility of deep sea CO₂ release experiments in understanding the biology of a high-CO₂ ocean: effects of hypercapnia on deep sea meiofauna. *J. Geophys. Res. C Oceans* 110:1–18.
- Beniash, E., A. Ivanina, N. S. Lieb, I. Kurochkin & I. M. Sokolova. 2010. Elevated level of carbon dioxide affects metabolism and shell formation in oysters *Crassostrea virginica*. *Mar. Ecol. Prog. Ser.* 419:95–108.
- Bookhout, C. G. & J. D. J. Costlow. 1970. Nutritional effects of *Artemia* from different locations on larval development of crabs. *Helgol. Wiss. Meeresunters.* 20:435–442.
- Byrne, M. 2011. Impact of ocean warming and ocean acidification on marine invertebrate life history stages: vulnerabilities and potential for persistence in a changing ocean. *Oceanogr. Mar. Biol. Annu. Rev.* 49:1–42.
- Caldeira, K. & M. E. Wickett. 2003. Oceanography: anthropogenic carbon and ocean pH. *Nature* 425:365.
- Cameron, J. N. 1978. Effects of hypercapnia on blood acid-base status, NaCl fluxes, and trans-gill potential in freshwater blue crabs, *Callinectes sapidus*. *J. Comp. Physiol. B* 123:137–141.
- Cameron, J. N. & G. K. Iwama. 1987. Compensation of progressive hypercapnia in channel catfish and blue crabs. *J. Exp. Biol.* 197:183–197.
- Costlow, J. D. J. & C. G. Bookhout. 1959. The larval development of *Callinectes sapidus* Rathbun reared in the laboratory. *Biol. Bull.* 116:373–396.
- Dodd, L. F., J. H. Grabowski, M. F. Piehler, I. Westfield & J. B. Ries. 2015. Ocean acidification impairs crab foraging behaviour. *Proc. Biol. Sci.* 282:20150333.
- Doney, S. C., V. J. Fabry, R. A. Feely & J. A. Kleypas. 2009. Ocean acidification: the other CO₂ problem. *Annu. Rev. Mar. Sci.* 1:169–192.
- Duarte, C. M. C. 2007. Marine ecology warms up to theory. *Trends Ecol. Evol.* 22:331–333.
- Dupont, S., N. Dorey & M. Thorndyke. 2010. What meta-analysis can tell us about vulnerability of marine biodiversity to ocean acidification? *Estuar. Coast. Shelf Sci.* 89:182–185.
- Dupont, S., J. Havenhand, W. Thorndyke, L. Peck & M. Thorndyke. 2008. Near-future level of CO₂-driven ocean acidification radically

- affects larval survival and development in the brittlestar *Ophiothrix fragilis*. *Mar. Ecol. Prog. Ser.* 373:285–294.
- Dupont, S. & M. C. Thorndyke. 2009. Impact of CO₂-driven ocean acidification on invertebrates early life-history – what we know, what we need to know and what we can do. *Biogeosciences* 6:3109–3131.
- Eggleston, D. B., E. G. Johnson, G. T. Kellison, G. R. Plaia & C. L. Huggett. 2008. Pilot evaluation of early juvenile blue crab stock enhancement using a replicated BACI design. *Rev. Fish. Sci.* 16:91–100.
- Egilsdottir, H., J. I. Spicer & S. D. Rundle. 2009. The effect of CO₂ acidified sea water and reduced salinity on aspects of the embryonic development of the amphipod *Echinogammarus marinus* (Leach). *Mar. Pollut. Bull.* 58:1187–1191.
- Fabry, V. J., B. A. Seibel, R. A. Feely & J. C. Orr. 2008. Impacts of ocean acidification on marine fauna and ecosystem processes. *ICES J. Mar. Sci.* 65:414–432.
- Feely, R. A., S. C. Doney & S. R. Cooley. 2009. Ocean acidification: present conditions and future changes in a high-CO₂ world. *Oceanography* 22:36–47.
- Feely, R. A., C. L. Sabine, K. Lee, W. Berelson, J. Kleypas, V. J. Fabry & F. J. Millero. 2004. Impact of anthropogenic CO₂ on the CaCO₃ system in the oceans. *Science* 305:362–366.
- Glandon, H. L. & T. J. Miller. 2016. No effect of high pCO₂ on juvenile blue crab, *Callinectes sapidus*, growth and consumption despite positive responses to concurrent warming. *ICES J. Mar. Sci.* 74: 1201–1209.
- Gooding, R. A., C. D. G. Harley & E. Tang. 2009. Elevated water temperature and carbon dioxide concentration increase the growth of a keystone echinoderm. *Proc. Natl. Acad. Sci. USA* 106:9316–9321.
- Hall-Spencer, J., M. Thorndyke & S. Dupont. 2015. Impact of ocean acidification on marine organisms—unifying principles and new paradigms. *Water* 7:5592–5598.
- IPCC. 2007. Climate change 2007: the physical science basis. In Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change, Cambridge University.
- Kurihara, H. & Y. Shirayama. 2004. Effects of increased atmospheric CO₂ on sea urchin early development. *Mar. Ecol. Prog. Ser.* 274:161–169.
- Long, W. C., K. M. Swiney, C. Harris, H. N. Page & R. J. Foy. 2013. Effects of ocean acidification on juvenile red king crab (*Paralithodes camtschaticus*) and tanner crab (*Chionoecetes bairdi*) growth, condition, calcification, and survival. *PLoS One* 8: e60959.
- Malakoff, D. 2012. Researchers struggle to assess responses to ocean acidification. *Science* 338:27–28.
- McLaskey, A. J. K., P. McElhany, M. B. Olson, S. Busch, M. Maher & A. K. Winans. 2016. Development of *Euphausia pacifica* (krill) larvae is impaired under pCO₂ levels that are currently observed in the northeast Pacific. *Mar. Ecol. Prog. Ser.* 555:65–78.
- Neufeld, D. S. & J. N. Cameron. 1992. Postmoult uptake of calcium by the blue crab (*Callinectes sapidus*) in water of low salinity. *J. Exp. Biol.* 171:283–299.
- O'Connor, M. I., J. F. Bruno, S. D. Gaines, B. S. Halpern, S. E. Lester, B. P. Kinlan & J. M. Weiss. 2007. Temperature control of larval dispersal and the implications for marine ecology, evolution, and conservation. *Proc. Natl. Acad. Sci. USA* 104:1266–1271.
- Orr, J. C., V. J. Fabry, O. Aumont, L. Bopp, S. C. Doney, R. A. Feely, A. Gnanadesikan, N. Gruber, A. Ishida, F. Joos, R. M. Key, K. Lindsay, E. Maier-Reimer, R. Matear, P. Monfray, A. Mouchet, R. G. Najjar, G.-K. Plattner, K. B. Rodgers, C. L. Sabine, J. L. Sarmiento, R. Schlitzer, R. D. Slater, I. J. Totterdell, M.-F. Weirig, Y. Yamanaka & A. Yool. 2005. Anthropogenic ocean acidification over the twenty-first century and its impact on calcifying organisms. *Nature* 437:681–686.
- Pechenik, J. A. 1999. On the advantages and disadvantages of larval stages in benthic marine invertebrate life cycles. *Mar. Ecol. Prog. Ser.* 177:269–297.
- Pespeni, M. H., E. Sanford, B. Gaylord, T. M. Hill, J. D. Hosfelt, H. K. Jaris, M. LaVigne, E. A. Lenz, A. D. Russell, M. K. Young & S. R. Palumbi. 2013. Evolutionary change during experimental ocean acidification. *Proc. Natl. Acad. Sci. USA* 110:1–6.
- Przeslawski, R., S. Ah Yong, M. Byrne, G. Wörheides & P. Hutchings. 2008. Beyond corals and fish: the effects of climate change on noncoral benthic invertebrates of tropical reefs. *Glob. Change Biol.* 14:2773–2795.
- Raven, J., K. Caldeira, H. Elderfield, O. Hoegh-Guldberg, P. Liss, U. Riebesell, J. Shepherd, C. Turley & A. Watson. 2005. Ocean acidification due to increasing atmospheric carbon dioxide. The Royal Society policy document 12/05. Clyvedon Press, Cardiff, UK.
- Riebesell, U. 2008. Climate change: acid test for marine biodiversity. *Nature* 454:46–47.
- Ries, J. B., L. Cohen & D. C. McCorkle. 2009. Marine calcifiers exhibit mixed responses to CO₂-induced ocean acidification. *Geology* 37:1131–1134.
- Sabine, C. L., R. A. Feely, N. Gruber, R. M. Key, K. Lee, J. L. Bullister, R. Wanninkhof, C. S. Wong, D. W. R. Wallace, B. Tilbrook, F. J. Millero, T.-H. Peng, A. Kozyr, T. Ono & A. F. Rios. 2004. The oceanic sink for anthropogenic CO₂. *Science* 305:367–371.
- Shanks, A. L., B. A. Grantham & M. H. Carr. 2003. Propagule dispersal distance and the size and spacing of marine reserves. *Ecol. Appl.* 13: S159–S169.
- Silliman, B. R. & M. D. Bertness. 2002. A trophic cascade regulates salt marsh primary production. *Proc. Natl. Acad. Sci. USA* 99:10500–10505.
- Small, D. P., P. Calosi, D. Boothroyd, S. Widdicombe & J. I. Spicer. 2016. The sensitivity of the early benthic juvenile stage of the European lobster *Homarus gammarus* (L.) to elevated pCO₂ and temperature. *Mar. Biol.* 163:1–12.
- Sulkin, S. D. 1975. The significance of diet in the growth and development of larvae of the blue crab, *Callinectes sapidus* Rathbun, under laboratory conditions. *J. Exp. Mar. Biol. Ecol.* 20:119–135.
- Sulkin, S. D. & C. E. Epifanio. 1975. Comparison of rotifers and other diets for rearing early larvae of the blue crab, *Callinectes sapidus* Rathbun. *Estuar. Coast. Mar. Sci.* 3:109–113.
- Takahashi, T. 2004. Ocean science. The fate of industrial carbon dioxide. *Science* 305:352–353.
- Vigh, D. & J. Dendinger. 1982. Temporal relationships of postmolt deposition of calcium, magnesium, chitin and protein in the cuticle of the Atlantic blue crab, *Callinectes sapidus* Rathbun. *Comp. Biochem. Physiol.* 72:365–369.
- Wahl, M., V. Saderne & Y. Sawall. 2016. How good are we at assessing the impact of ocean acidification in coastal systems? Limitations, omissions and strengths of commonly used experimental approaches with special emphasis on the neglected role of fluctuations. *Mar. Freshwater Res.* 67:25–36.
- Walther, K., K. Anger & H. O. Pörtner. 2010. Effects of ocean acidification and warming on the larval development of the spider crab *Hyas araneus* from different latitudes (54° vs. 79°N). *Mar. Ecol. Prog. Ser.* 417:159–170.
- Weydmann, A., J. E. Søreide, S. Kwasniewski & S. Widdicombe. 2012. Influence of CO₂-induced acidification on the reproduction of a key Arctic copepod *Calanus glacialis*. *J. Exp. Mar. Biol. Ecol.* 428:39–42.
- Whiteley, N. 2011. Physiological and ecological responses of crustaceans to ocean acidification. *Mar. Ecol. Prog. Ser.* 430:257–271.
- Wood, H. L., S. P. Eriksson, M. Nordborg & H. K. Styf. 2015. The effect of environmental stressors on the early development of the Norway lobster *Nephrops norvegicus* (L.). *J. Exp. Mar. Biol. Ecol.* 473:35–42.
- Zmora, O., A. Findiesen, J. Stubblefield, V. Frenkel & Y. Zohar. 2005. Large-scale juvenile production of the blue crab *Callinectes sapidus*. *Aquaculture* 244:129–139.